



HABIB UNIVERSITY

DHANANI SCHOOL OF SCIENCE AND ENGINEERING

EE251/CE252	Spring'26	Quiz 2 SOLUTION	Marks 30 Weight 4%	Feb 25, 2026
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Important Notes

Clearly write your name and student ID on top.

CLO-2	Calculate response of DT and CT LTI systems to various inputs and categorize LTI systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
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Problem 1 [10 pts]

Consider a DT LTI system with impulse response

$$h[n] = 3\delta[n-3] + 2(\delta[n-2] + \delta[n-4]) + \delta[n-1] + \delta[n-5]$$

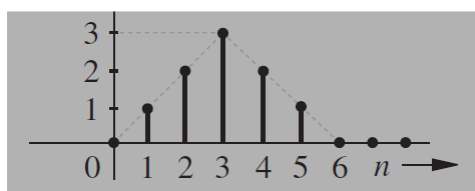
Find its output to the input

$$x[n] = \sum_{k=0}^5 \delta[n-k]$$

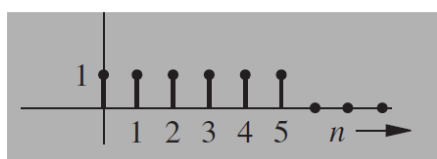
Solution:

We first notice that the input and impulse response form following sequences

$h[n]$:

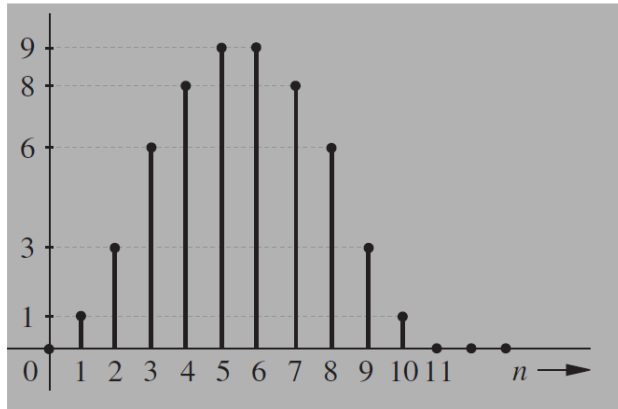


$x[n]$:



Since systems is LTI, output is simply $y[n] = x[n] * h[n]$. Using any of the methods of convolution (sliding-tape would be fastest in this case), we find the output to be

$y[n]$:



Problem 2 [10 pts]

Consider the DT system described by $y[n] = A^{N_0}x[n - N_0]$, where $x[n]$ is the input, $y[n]$ the output, and N_0, A are constants. Determine if this system is LTI.

Solution:

System is linear, since

$$S\{a_1x_1[n] + a_2x_2[n]\} = A^{N_0}(a_1x_1[n - N_0] + a_2x_2[n - N_0]) = a_1A^{N_0}x_1[n - N_0] + a_2A^{N_0}x_2[n - N_0] = a_1S\{x_1[n]\} + a_2S\{x_2[n]\}.$$

Let's check for TI

$$y_1[n] = S\{x[n - N]\} = A^{N_0}x[n - N - N_0]$$

And

$$y[n - N] = A^{N_0}x[n - N - N_0]$$

Clearly $y_1[n] = y[n - N]$. The system is thus TI also.

In conclusion, the given system is LTI.

Problem 3 [10 pts]

Consider a DT LTI system with impulse response

$$h[n] = 4(0.8)^n u[n] + 2\delta[n + 1]$$

Determine if the system is memoryless, causal, and BIBO stable.

Solution:

The given impulse response exhibits an exponential decay for $n \geq 0$ and has a value of 2 at $n = -1$.

System is not memoryless since it violates $h[n] = 0, n \neq 0$ condition.

System is not causal since $h[-1] = 2$ (violating $h[n] = 0$ for $n < 0$ condition).

System is BIBO stable since $\sum_n |h[n]| < \infty$ (easy to show as $h[n]$ comprises of a 2 and a decaying geometric series with $|r| = 0.8 < 1$, which we know are convergent).